FeedbackIQ — Multi-Domain Customer Feedback Intelligence

Sentiment · complaint taxonomy · retrieval-augmented Q&A over 642,692 reviews

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github.com/<your-username>/FeedbackAnalytics_LLM

The common claim is that fine-tuning buys accuracy. Across Amazon, Yelp and Twitter feedback, it mostly bought **consistency** — a 2-point aggregate edge over logistic regression that hides a much larger story once broken down by platform. Built end-to-end: three-stage NLP pipeline, RAG chatbot, full evaluation framework, containerised and deployment-ready.

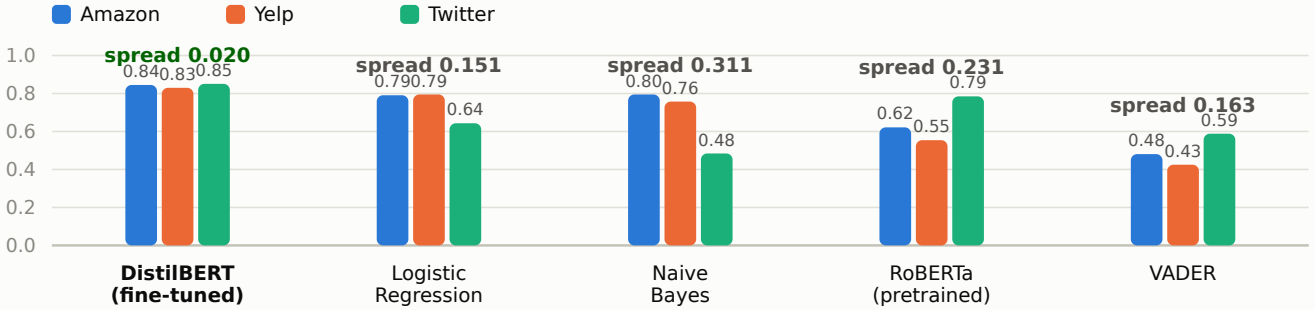
HEADLINE RESULT

0.8244
Macro F1, fine-tuned DistilBERT (5-model comparison)

0.020
Cross-platform spread — vs. 0.15–0.31 for every baseline

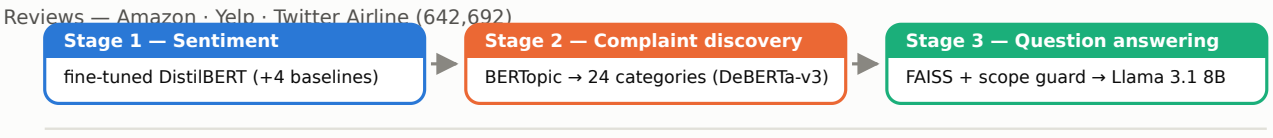
$p=1.7\times10^{-21}$
Significance vs. logistic regression (McNemar's exact test)

642,692
Reviews — Amazon, Yelp, Twitter Airline, one shared eval set



Macro F1 by platform, five models. The aggregate score hides this: every baseline swings 15–31 points depending which platform it's handed — fine-tuning bought a model that doesn't need to know in advance which one it is.

SYSTEM ARCHITECTURE



Two containers — FastAPI backend, Streamlit frontend. A refused question returns in 112ms.

RETRIEVAL & TAXONOMY

RAG vs. standalone LLM — the honest trade-off

Retrieval: **Precision@3 0.90** at **21.7ms** mean latency over the full 642k-review index.
Grounded generation scored **lower** on answer relevancy (0.61 vs 0.75) than the standalone LLM — but it's the only one whose answers trace to actual evidence (faithfulness 0.54, unmeasurable for the ungrounded baseline). Reported as a trade-off, not a win.
Metadata-tagging ablation: **+0.118** faithfulness, **−0.202** relevancy — the one design decision in the system that was measured rather than argued.

Complaint taxonomy — BERTopic → 24 categories

121 raw topics (per-platform BERTopic + UMAP + HDBSCAN) consolidated to a 24-category cross-domain taxonomy, assigned via DeBERTa-v3 zero-shot entailment.
Held-out evaluation (n=450): mean coherence **0.633**, **27.6%** unclassified — reported rather than force-fit, on the same design philosophy as the retrieval scope guard.
Amazon alone produced **69** of the 121 raw topics from a third of the documents — semantic breadth, not a tuning artefact.

ENGINEERING

Reproducibility

Seeds fixed at 42 throughout. Every table in the results cites the exact file that produced it. Classical training and evaluation tracked in MLflow. Dependencies pinned.

Service

FastAPI backend + Streamlit frontend, two Docker images, API-key auth, health checks, GCP deployment runbook (Compute Engine, docker-compose). `x-api-key` required on every route except `/health`.

LIMITATIONS — STATED PLAINLY, BECAUSE THEY BOUND WHAT THE NUMBERS MEAN

- **Labels come from star ratings** — a 4-star review containing a complaint is labelled positive, bounding every model's ceiling.
- **Generation evaluation is n=5**, same model family generating and grading — directional, not precise, and reported as such.
- **The similarity threshold (0.35) is unvalidated** — a starting point, not a calibrated constant; the cheapest outstanding experiment.
- **3 of 4 planned ablations weren't run** — since retrieval relevance is the identified bottleneck, those are the most consequential gaps left.

Python 3.12 · PyTorch · Transformers · sentence-transformers · FAISS · LangChain · Groq · BERTopic · UMAP · HDBSCAN · scikit-learn · spaCy · RAGAS · MLflow · FastAPI · Streamlit · Docker

Full results & code on GitHub →